3.10 Formal Power Series and Laurent Series

Definition 3.10.1 (Formal Power Series)
*Let A be a commutative ring. Define the **formal power series ring** $A[[X]]$ to be the set of sequences $(c_n)_{n\geq 0}$ in A, with componentwise addition and convolution product, written as formal sums*

$$f(X) := \sum_{n=0}^{\infty} c_n X^n$$

so $A[X] \subset A[[X]]$ is the subring of finitely-supported sequences. If A is an integral domain, so is $A[[X]]$, with $v(fg) = v(f) + v(g)$ for $v(f) := \min\{n \mid c_n \neq 0\}$ (the coefficient of $X^{v(f)+v(g)}$ in fg is $c_{v(f)}d_{v(g)} \neq 0$, every other contributing pair vanishing).

Proposition 3.10.2 (Units of a Formal Power Series Ring)
Let A be a commutative ring and $u \in A[[X]]$, with constant coefficient u_0 (the coefficient of X^0). If $u_0 \in A^\star$ then u is a unit of $A[[X]]$. When $A = k$ is a field, the converse holds too, so $u \in k[[X]]^\star \iff u_0 \neq 0$.

Proof. Suppose $u_0 \in A^\star$. Set $w := 1 - u_0^{-1}u$, so w has constant coefficient 0; then $u_0^{-1}\sum_{i\geq 0} w^i$ is a well-defined element of $A[[X]]$ (at each order n, only finitely many terms w^i contribute, as $v(w) \geq 1$), and $u \cdot u_0^{-1}\sum_{i\geq 0} w^i = (1 - w)\sum_{i\geq 0} w^i = 1$ (telescoping), an explicit inverse of u.

For the converse when $A = k$ is a field: if u is a unit with inverse v, then $u_0 v_0 = (uv)_0 = 1$, so $u_0 \neq 0$; conversely $u_0 \neq 0$ in the field k gives $u_0 \in k^\star$, so the first paragraph applies. ∎

Lemma 3.10.3 (e-th Roots of Units in a Formal Power Series Ring)
Let k be a field, $e \geq 1$ an integer invertible in k (i.e. $\mathrm{char}(k) \nmid e$), and $u \in k[[X]]^\star$. If $u(0)$ has an e-th root in k (automatic if k is algebraically closed), then u has an e-th root in $k[[X]]^\star$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Let $\lambda \in k$ with $\lambda^e = u(0)$; replacing u by $\lambda^{-e}u$ (whose e-th root, if found, gives one for u after multiplying by λ), we may assume $u(0) = 1$. Construct $y = 1 + \sum_{n\geq 1} y_n X^n$ recursively: set $y^{(0)} := 1$, and given $y^{(n-1)} := 1 + \sum_{j=1}^{n-1} y_j X^j$ with $\left(y^{(n-1)}\right)^e \equiv u \pmod{X^n}$ (trivial for $n = 1$, as $u(0) = 1$), write $\left(y^{(n-1)}\right)^e = u + c_n X^n + O(X^{n+1})$ for some $c_n \in k$; by the binomial theorem,

$$\left(y^{(n-1)} + y_n X^n\right)^e \equiv \left(y^{(n-1)}\right)^e + e\left(y^{(n-1)}\right)^{e-1} y_n X^n \pmod{X^{n+1}}$$

(every other term has order $\geq 2n \geq n + 1$), and $\left(y^{(n-1)}\right)^{e-1} \equiv 1 \pmod{X}$ (as $y^{(n-1)}(0) = 1$), so the coefficient of X^n on the right is $c_n + ey_n$. As e is invertible in k, setting $y_n := -c_n/e$ gives $y^{(n)} := y^{(n-1)} + y_n X^n$ with $\left(y^{(n)}\right)^e \equiv u \pmod{X^{n+1}}$, continuing the induction. The resulting $y := 1 + \sum_{n\geq 1} y_n X^n \in k[[X]]$ satisfies $y^e = u$ (agreeing with u to every order), and is a unit as $y(0) = 1 \neq 0$ (3.10.2). ∎

Definition 3.10.4 (Formal Laurent Series)
*Let A be a commutative ring. Define the **formal Laurent series ring** $A((X))$ to be the set of sequences $(c_n)_{n\in\mathbb{Z}}$ in A bounded below, with the same convolution product; then $A[[X]] \subset A((X))$ is the subring with $c_n = 0$ for $n < 0$. Write $v(h) := \min\{n \mid c_n \neq 0\}$ for $0 \neq h = \sum_n c_n X^n \in A((X))$, and $v(0) := \infty$.*

Proposition 3.10.5 (Summable Families in $A((X))$)
Let A be a commutative ring and $(h_i)_{i\in I}$ a family in $A((X))$ such that, for every $N \in \mathbb{Z}$, all but finitely many $i \in I$ satisfy $v(h_i) \geq N$. Then

$$\sum_{i\in I} h_i \in A((X))$$

is well-defined: for every $m \in \mathbb{Z}$, only finitely many $i \in I$ have nonzero coefficient of X^m in h_i, and the resulting sequence of coefficients (of $\sum_i h_i$) is bounded below. Moreover, derivatives may be computed termwise,

$$\left(\sum_{i\in I} h_i\right)' = \sum_{i\in I} h_{i'}$$

and multiplication distributes over the sum: for any $g \in A((X))$, $(gh_i)_{i\in I}$ also meets the hypothesis, and

$$g \cdot \sum_{i\in I} h_i = \sum_{i\in I} gh_i$$

Proof. **Bounded below.** Taking $N := 0$, all but finitely many i have $v(h_i) \geq 0$, that is $h_i \in A[[X]]$; the remaining finitely many h_i are each individually bounded below (as elements of $A((X))$), so $\sum_i h_i$ is bounded below by the minimum of 0 and these finitely many bounds.

Local finiteness. Fix $m \in \mathbb{Z}$ and take $N := m+1$: all but finitely many i have $v(h_i) \geq m+1 > m$, that is coefficient of X^m in h_i equal to 0; so only the finitely many remaining i can contribute a nonzero coefficient of X^m, giving a well-defined finite sum $\sum_i (h_i)_m \in A$ for the coefficient of X^m in $\sum_i h_i$.

Termwise derivatives. Since $v(h_{i'}) \geq v(h_i) - 1$ (differentiation shifts a coefficient at index n to index $n-1$, scaled by n; it strictly cannot lower the least nonzero index by more than one), $(h_{i'})_{i \in I}$ also meets the hypothesis (for every N, all but finitely many i have $v(h_i) \geq N+1$, hence $v(h_{i'}) \geq N$), so $\sum_i h_{i'}$ is itself well-defined by the case just proved. Fix m; by local finiteness applied to degree $m+1$, a finite set $I_0 \subset I$ contains every i with nonzero coefficient of X^{m+1} in h_i – in particular every i that can contribute to the coefficient of X^m in either $\left(\sum_i h_i\right)'$ or $\sum_i h_{i'}$, both computed from the coefficient of X^{m+1} in the h_i's. Restricted to the finite set I_0, differentiation and summation commute (both reduce to finite operations on coefficients), so the two coefficients of X^m agree; as m was arbitrary, $\left(\sum_i h_i\right)' = \sum_i h_{i'}$.

Distributivity. If $g = 0$ both sides are 0; otherwise $v(gh_i) \geq v(g) + v(h_i)$ for every i (every coefficient of gh_i below index $v(g) + v(h_i)$ vanishes, as any pair contributing to it has an index below $v(g)$ or below $v(h_i)$, where g, resp. h_i, already vanishes). So, given N, taking $N' := N - v(g)$ in the hypothesis on (h_i), all but finitely many i have $v(h_i) \geq N'$, hence $v(gh_i) \geq v(g) + N' = N$: the family $(gh_i)_{i \in I}$ meets the hypothesis too. For the equality, fix m; by local finiteness (applied to (h_i), at every degree q with $g_{m-q} \neq 0$ and $m-q$ within g's finitely-many-relevant range, itself finite as convolution $(g \cdot H)_m$ is already a finite sum for any $H \in A((X))$, (3.10.1)) only finitely many pairs (i, q) have $g_{m-q}(h_i)_q \neq 0$; both $\left(g \cdot \sum_i h_i\right)_m = \sum_q g_{m-q}\left(\sum_i h_i\right)_q = \sum_q g_{m-q} \sum_i (h_i)_q$ and $\left(\sum_i gh_i\right)_m = \sum_i (gh_i)_m = \sum_i \sum_q g_{m-q}(h_i)_q$ reduce to the same finite double sum over these pairs (i, q), so agree; as m was arbitrary, $g \cdot \sum_i h_i = \sum_i gh_i$. ∎

Proposition 3.10.6 ($k((X))$ is a Field of Fractions for $k[[X]]$)
Let k be a field. Then $k((X))$ is a field, and every element is a/b for $a, b \in k[[X]]$, $b \neq 0$ – that is, $k((X))$ is a field of fractions for the subring $k[[X]] \subset k((X))$. Moreover, the order function $v : k((X))^\star \to \mathbb{Z}$ is a group homomorphism under multiplication: $v(fg) = v(f) + v(g)$, and $v(f^{-1}) = -v(f)$.

Proof. Every $c \neq 0$ in $k((X))$ is $c = X^{n_0}u$ for n_0 its least nonzero index and $u := X^{-n_0}c \in k[[X]]$ with nonzero constant coefficient; by (3.10.2), u is a unit of $k[[X]]$, giving an explicit inverse $X^{-n_0}u^{-1}$ of c – so $k((X))$ is a field. This same decomposition writes c as u/X^{-n_0} if $n_0 < 0$, or as $uX^{n_0} \in k[[X]]$ itself if $n_0 \geq 0$ – in either case a ratio of two elements of $k[[X]]$.

For $0 \neq f, g \in k((X))$, write $f = X^{n_0}u$, $g = X^{m_0}w$ as above ($u, w \in k[[X]]^\star$); then $fg = X^{n_0+m_0}uw$ with $uw \in k[[X]]^\star$ again (nonzero constant coefficient, the product of those of u and w), so $v(fg) = n_0 + m_0 = v(f) + v(g)$. Applying this to f, f^{-1}: $0 = v(1) = v(ff^{-1}) = v(f) + v(f^{-1})$, giving $v(f^{-1}) = -v(f)$. ∎

Proposition 3.10.7 (Formal Power Series Ring is a DVR)
Let k be a field. Then $k[[X]]$ is a discrete valuation ring with fraction field $k((X))$ (3.10.6), maximal ideal (X), and valuation the order function v (3.10.1), (3.10.4).

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$k[[X]]$ is a domain ((3.10.1), k a domain) and not a field, as $X \neq 0$ has no inverse ($v(Xg) = 1 + v(g) \neq 0$ for every $g \in k[[X]]$). Given $0 \neq a \in k[[X]]$ with $n_0 := v(a) \geq 0$, write $a = X^{n_0}u$ for $u := X^{-n_0}a \in k[[X]]$ (well-defined as $n_0 \geq 0$), which has nonzero constant coefficient, so $u \in k[[X]]^\star$ (3.10.2): this is exactly the uniformised-element criterion (3.31.2)e), with $\pi := X$. ∎

Definition 3.10.8 (Formal Derivative)
Let A be a commutative ring. For $f = \sum_n c_n X^n \in A((X))$, define $f' := \sum_n n c_n X^{n-1} \in A((X))$.

Proposition 3.10.9 (Product Rule)
Let A be a commutative ring and $f, g \in A((X))$. Then $(fg)' = f'g + fg'$.

Proof. Write $f = \sum_i a_i X^i$, $g = \sum_j b_j X^j$. The coefficient of X^{n-1} in $(fg)'$ is n times the coefficient of X^n in fg, that is $n \sum_{i+j=n} a_i b_j$. The coefficient of X^{n-1} in $f'g$ is $\sum_{i+j=n} (ia_i) b_j$ (as f' has coefficient ia_i at degree $i-1$), and in fg' is $\sum_{i+j=n} a_i (jb_j)$. Summing,

$$\sum_{i+j=n} ia_ib_j + \sum_{i+j=n} a_ijb_j = \sum_{i+j=n} (i+j)a_ib_j = n\sum_{i+j=n} a_ib_j$$

matching the coefficient of X^{n-1} in $(fg)'$. As n was arbitrary, $(fg)' = f'g + fg'$. ∎

Lemma 3.10.10 (Power Rule)
Let A be a commutative ring and $\psi \in A((X))$. Then $(\psi^n)' = n\psi^{n-1}\psi'$ for every $n \geq 0$. If moreover ψ is a unit of $A((X))$, this also holds for every $n < 0$.

Proof. **$n \geq 0$.** Induction on n: the case $n = 0$ is $0 = 0$ ($\psi^0 = 1$ is constant). Given $(\psi^n)' = n\psi^{n-1}\psi'$, the product rule (3.10.9) gives

$$(\psi^{n+1})' = (\psi \cdot \psi^n)' = \psi'\psi^n + \psi(\psi^n)' = \psi'\psi^n + n\psi^n\psi' = (n+1)\psi^n\psi'$$

$n < 0$, ψ a unit. Differentiating $\psi\psi^{-1} = 1$ with the product rule, $0 = \psi'\psi^{-1} + \psi(\psi^{-1})'$, so $(\psi^{-1})' = -\psi^{-2}\psi'$ – the case $n = -1$. For $n \leq -2$, write $n = -r$, $r \geq 2$: applying the $n \geq 0$ case just shown to the element $\psi^{-1} \in A((X))$ (no hypothesis on its order was used there),

$$(\psi^{-r})' = \left((\psi^{-1})^r\right)' = r(\psi^{-1})^{r-1}(\psi^{-1})' = r(\psi^{-1})^{r-1}(-\psi^{-2}\psi') = -r\psi^{-r-1}\psi' = n\psi^{n-1}\psi'$$

∎

Proposition 3.10.11 (Substitution into a Formal Power Series)
Let A be a commutative ring, $f = \sum_{n\geq 0} c_nX^n \in A[[X]]$, and $\psi \in XA[[X]]$ (that is, $\psi(0) = 0$). Then

$$(f \circ \psi)(X) := \sum_{n\geq 0} c_n\psi(X)^n$$

is a well-defined element of $A[[X]]$.

Proof. $\psi \in XA[[X]]$ means $\psi^n \in X^nA[[X]]$ for every $n \geq 0$ (a product of n elements of $XA[[X]]$), that is $v(\psi^n) \geq n$. So for every N, all $n \geq N$ satisfy $v(c_n\psi^n) \geq n \geq N$: the family $(c_n\psi^n)_{n\geq 0}$ meets (3.10.5)'s hypothesis (only $n < N$, finitely many, can fail it), giving $\sum_n c_n\psi^n \in A[[X]]$ (bounded below by 0, as every term already is). ∎

Proposition 3.10.12 (Chain Rule, Power Series)
In the setting of (3.10.11), $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$.

Proof. By the power rule (3.10.10), $f \circ \psi = \sum_n c_n\psi^n$ is a sum of the summable family $(c_n\psi^n)_{n\geq 0}$ ((3.10.11)'s proof), each term of which satisfies $(c_n\psi^n)' = c_n(\psi^n)'$ (linearity of $(-)'$, immediate from (3.10.8)'s termwise formula, as $c_n \in A$ is a constant). So, by termwise differentiation (3.10.5),

$$(f \circ \psi)' = \sum_{n\geq 0} (c_n\psi^n)' = \sum_{n\geq 0} c_n(\psi^n)' = \sum_{n\geq 0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n\geq 0}$ meets (3.10.5)'s hypothesis by the same order bound as $(c_n\psi^n)_{n\geq 0}$ ((3.10.11)'s proof, shifted down one degree), so distributivity (same reference) gives

$$\sum_{n\geq 0} nc_n\psi^{n-1}\psi' = \left(\sum_{n\geq 0} nc_n\psi^{n-1}\right)\psi'$$

Reindexing $m = n - 1$, $\sum_{n\geq 0} nc_n\psi^{n-1} = \sum_{m\geq 0}(m+1)c_{m+1}\psi^m = f' \circ \psi$, giving $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$. ∎

Proposition 3.10.13 (Substitution into a Formal Laurent Series)
Let k be a field, $f = \sum_{n\geq n_0} c_nX^n \in k((X))$, and $\psi = Xv \in Xk[[X]]$ for a unit $v \in k[[X]]^\star$ (that is, ψ has invertible linear term, (3.10.2)). Then

$$(f \circ \psi)(X) := \sum_{n\geq n_0} c_n\psi(X)^n$$

is a well-defined element of $k((X))$ (with ψ^n, $n < 0$, the power of ψ in the field $k((X))$, (3.10.6)).

Proof. As v is a group homomorphism under multiplication (3.10.6), $v(\psi^n) = n \cdot v(\psi) = n$ for every $n \in \mathbb{Z}$, as $v(\psi) = 1$. Given N, every $n \geq \max(N, n_0)$ then satisfies $v(c_n\psi^n) \geq n \geq N$: the family $(c_n\psi^n)_{n\geq n_0}$ meets (3.10.5)'s hypothesis (only $n_0 \leq n < N$, finitely many, can fail it), giving $\sum_n c_n\psi^n \in k((X))$, bounded below by n_0 (as every term already is). ∎

Proposition 3.10.14 (Chain Rule, Laurent Series)
In the setting of (3.10.13), $(f\circ\psi)' = (f'\circ\psi)\cdot\psi'$.

Proof. By the power rule ((3.10.10), applicable for every $n\in\mathbb{Z}$ as ψ is a unit of $k((X))$), $f\circ\psi = \sum_{n\geq n_0} c_n\psi^n$ is a sum of the summable family $(c_n\psi^n)_{n\geq n_0}$ ((3.10.13)'s proof), each term of which satisfies $(c_n\psi^n)' = c_n(\psi^n)'$ (linearity of $(-)'$, as in (3.10.12)'s proof). So, by termwise differentiation (3.10.5),

$$(f\circ\psi)' = \sum_{n\geq n_0}(c_n\psi^n)' = \sum_{n\geq n_0} c_n(\psi^n)' = \sum_{n\geq n_0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n\geq n_0}$ meets (3.10.5)'s hypothesis by the same order bound as $(c_n\psi^n)_{n\geq n_0}$ ((3.10.13)'s proof, shifted down one degree), so distributivity (same reference) gives

$$\sum_{n\geq n_0} nc_n\psi^{n-1}\psi' = \left(\sum_{n\geq n_0} nc_n\psi^{n-1}\right)\psi' = (f'\circ\psi)\cdot\psi'$$

■

Lemma 3.10.15 (Substitution is Multiplicative)
In the setting of (3.10.13), for $f,g\in k((X))$, $(fg)\circ\psi = (f\circ\psi)(g\circ\psi)$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $f = \sum_{i\geq i_0} c_iX^i$, $g = \sum_{j\geq j_0} d_jX^j$, so $f\circ\psi = \sum_i c_i\psi^i$ and $g\circ\psi = \sum_j d_j\psi^j$ (sums of the summable families of (3.10.13)'s proof).

The double family is summable. As $v(\psi^n) = n$ for every n ((3.10.13)'s proof), $v\big(c_id_j\psi^{i+j}\big)\geq i+j$; given N, all but finitely many pairs $(i,j)\in\mathbb{Z}_{\geq i_0}\times\mathbb{Z}_{\geq j_0}$ have $i+j\geq N$ (only the finitely many pairs with $i_0\leq i<N-j_0$ can fail this), so $\big(c_id_j\psi^{i+j}\big)_{(i,j)}$ meets (3.10.5)'s hypothesis and has a well-defined sum $\Sigma := \sum_{(i,j)} c_id_j\psi^{i+j}$.

$\Phi(f)\Phi(g) = \Sigma$. By distributivity ((3.10.5), applied twice), for each fixed j, $d_j\psi^j\cdot\sum_i c_i\psi^i = \sum_i c_id_j\psi^{i+j}$; fixing a target order m, only finitely many j have nonzero coefficient of X^m in $\sum_i c_id_j\psi^{i+j}$ (else infinitely many pairs (i,j) would contribute to the coefficient of X^m in Σ, contradicting its local finiteness), so $\sum_j\big(d_j\psi^j\sum_i c_i\psi^i\big) = \sum_j\sum_i c_id_j\psi^{i+j}$ is itself a well-defined sum (3.10.5) agreeing coefficientwise with Σ (both reduce, at each order m, to the same finite sum over the finitely many pairs (i,j) contributing there). The left side is $\sum_j d_j\psi^j\cdot(f\circ\psi) = (f\circ\psi)\sum_j d_j\psi^j = (f\circ\psi)(g\circ\psi)$ (distributivity again).

$\Sigma = (fg)\circ\psi$. $fg = \sum_N\Big(\sum_{i+j=N} c_id_j\Big)X^N$ (only finitely many pairs $i_0\leq i, j_0\leq j$ with $i+j=N$, namely $i_0\leq i\leq N-j_0$), so $(fg)\circ\psi = \sum_N\Big(\sum_{i+j=N} c_id_j\Big)\psi^N$. Fixing m, local finiteness of Σ (3.10.5) gives a finite set of pairs (i,j) contributing to the coefficient of X^m in Σ; grouping them by $N := i+j$ (finitely many values of N occur, each with finitely many pairs) shows the coefficient of X^m in Σ equals that in $\sum_N\Big(\sum_{i+j=N} c_id_j\Big)\psi^N = (fg)\circ\psi$. As m was arbitrary, $\Sigma = (fg)\circ\psi$.

Combining the two displayed equalities, $(f\circ\psi)(g\circ\psi) = \Sigma = (fg)\circ\psi$. ■

Lemma 3.10.16 (Substitution is Additive)
In the setting of (3.10.13), for $f,h\in k((X))$, $(f+h)\circ\psi = f\circ\psi + h\circ\psi$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $f = \sum_{n\geq n_0} c_nX^n$, $h = \sum_{n\geq m_0} d_nX^n$, and set $p_0 := \min(n_0,m_0)$, extending both coefficient sequences by $c_n := 0$ (resp. $d_n := 0$) for $p_0\leq n<n_0$ (resp. $<m_0$); this changes neither series. Then $f+h = \sum_{n\geq p_0}(c_n+d_n)X^n$, so $(f+h)\circ\psi = \sum_{n\geq p_0}(c_n+d_n)\psi^n$, and $f\circ\psi + h\circ\psi = \sum_{n\geq p_0} c_n\psi^n + \sum_{n\geq p_0} d_n\psi^n$, all three well-defined summable families (3.10.13).

Fix $m\in\mathbb{Z}$. As $v(\psi^n) = n$ for every $n\geq 0$ ((3.10.13)'s proof), only the finitely many indices n with $p_0\leq n\leq m$ can contribute a nonzero coefficient of X^m to any of the three sums above (each remaining $n>m$ satisfies $v(c_n\psi^n), v(d_n\psi^n), v((c_n+d_n)\psi^n)\geq n>m$); so the coefficient of X^m in each sum reduces to the same finite sum over $p_0\leq n\leq m$, and coefficientwise, writing γ_n for the coefficient of X^m in ψ^n, $(c_n+d_n)\gamma_n = c_n\gamma_n + d_n\gamma_n$. So the coefficients of X^m agree on both sides, for every m; hence $(f+h)\circ\psi = f\circ\psi + h\circ\psi$. ■